INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Pertuzumab + trastuzumab · with selected chemotherapy

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00350
Version 3

Treatment stages

01 **Before surgery · selected chemotherapy + antibodies**
21-day cycle

02 **After surgery · agreed adjuvant plan**
21-day cycle

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

SACTCHECK · CHAPTER 01

Understanding your treatment

Pertuzumab + trastuzumab · with selected chemotherapy · National Cancer Control Programme (NCCP) 00350

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathlessness, chest symptoms, severe diarrhoea, dehydration, jaundice, unusual bleeding or sudden illness promptly. Do not wait for the next planned test.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

What each medicine does

Treat HER2-positive early breast cancer as part of a plan intended to reduce the risk of it returning.

HER2-TARGETED ANTIBODY

Pertuzumab

HER2-targeted antibody



Blocks HER2 pairing with other receptors

Pertuzumab binds a different area of HER2 from trastuzumab. It limits receptor pairing, so the two antibodies act together at different points on the same growth pathway.

HER2 pairing → Pertuzumab blocks pairing → Growth signals reduced

What to watch for

Keep heart-scan appointments. Report diarrhoea, new breathlessness, swelling or a treatment reaction.

HER2-TARGETED ANTIBODY

Trastuzumab

HER2-targeted antibody



Targets HER2 growth signals

Trastuzumab binds to the HER2 protein on cells and helps reduce HER2-driven growth and recruit immune activity. HER2 testing identifies the cancers for which this approach is used.

HER2 on cell surface → Antibody binds → Growth signalling reduced

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

Two HER2-targeted antibodies may be combined with selected chemotherapy for higher-risk HER2-positive early breast cancer. Cancer extent, lymph nodes, hormone receptors, heart function and the chosen chemotherapy influence the plan. This card explains the antibody component; the companion chemotherapy needs its own regimen guide and assessment.

What could change the choice?

Ask which cancer features, earlier treatments and health factors led to this recommendation, which alternatives were considered and when the decision will be reviewed.

Your treatment and the next step

Schedule

intravenous (IV) pertuzumab and IV trastuzumab every 21 days. Before surgery: 3–6 cycles alongside the selected chemotherapy. After surgery: up to 18 cycles overall under the agreed adjuvant plan. If an anthracycline is used first, finish it before starting these antibodies with the taxane phase. Confirm the exact companion regimen and local access.

After this phase

The chosen chemotherapy, indication, response, tolerance and local access determine the next step. Do not treat this antibody framework as a complete chemotherapy protocol.

Confirm the selected plan

The source lists adjuvant pertuzumab reimbursement as not approved. Your team must confirm current access and the exact companion chemotherapy before this pathway is selected.

SACTCHECK · CHAPTER 02

Your schedule, tests and appointments

Pertuzumab + trastuzumab · with selected chemotherapy · NCCP 00350

The phases and how they connect

Treatment stages

01 **Before surgery · selected chemotherapy + antibodies**
21-day cycle

02 **After surgery · agreed adjuvant plan**
21-day cycle

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

21-DAY CYCLE
Before surgery · selected chemotherapy + antibodies

21-day cycle

Three to six antibody cycles with the selected chemotherapy. Anthracyclines, if used, finish before antibodies start with the taxane.

21-DAY CYCLE
After surgery · agreed adjuvant plan

21-day cycle

Up to 18 cycles overall under the source; confirm individual plan and current access. Chemotherapy is separately specified.

Relative sequence, not equal lengths or booked dates. Your team confirms actual appointments and the active medicines.

BASELINE → TREATMENT → FOLLOW-UP

Tests and milestones through treatment

Columns show sequence, not equal durations or appointments. A test is only complete when performed and reviewed. Echo timing and cancer-imaging timing are different. Symptoms can bring any check forward.

Check	Select indication / chemotherapy	Anthracycline first, if used	Antibodies with taxane	Surgery / adjuvant continuation	Completion / follow-up
Blood tests	Baseline antibody and companion-regimen checks	Follow the chosen anthracycline source	Before each cycle plus companion-regimen checks	Follow the current medicines	Follow abnormal results / next plan
Echo / cardiac assessment	Cardiac risk and baseline review	Reassess after anthracycline, before first HER2 dose	First echo 12 weeks after HER2 begins	Every 12 weeks during HER2; at completion	Follow-up for risk / chosen medicine
Cancer imaging / markers	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Reassess disease and treatment plan	Breast surveillance after early treatment
Surgery / radiotherapy	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Endocrine / bone health	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	dual-energy X-ray absorptiometry (bone density) (DEXA) when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00350 v3 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

Pertuzumab + trastuzumab · with selected chemotherapy: the whole pathway

HER2-positive early breast cancer · neoadjuvant or adjuvant framework

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Select indication / chemotherapy	Anthracycline first, if used	Antibodies with taxane	Surgery / adjuvant continuation	Completion / follow-up
Cancer medicines NCCP	Confirm diagnosis and exact regimen	See the dose-day map: antibodies with selected chemotherapy	Start antibodies with the taxane	Confirm indication, companion and continuation	Review benefit / tolerance and next plan
Surgery / radiotherapy EARLY	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Blood tests NCCP	Baseline antibody and companion-regimen checks	Follow the chosen anthracycline source	Before each cycle plus companion-regimen checks	Follow the current medicines	Follow abnormal results / next plan
Echo / cardiac checks NCCP · ESC	Cardiac risk and baseline review	Reassess after anthracycline, before first HER2 dose	First echo 12 weeks after HER2 begins	Every 12 weeks during HER2; at completion	Follow-up for risk / chosen medicine
Scans / tumour markers EARLY	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Reassess disease and treatment plan	Breast surveillance after early treatment
Endocrine / DEXA BONE	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	DEXA when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

What each check is for

1

Blood tests

full blood count (FBC), kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose. The selected chemotherapy can require extra or more frequent checks.

Check safety before the next dose.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

2

Heart tests

Check heart pumping function before starting HER2 treatment. Repeat echo or multigated acquisition (heart pumping) scan (MUGA) every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Monitor pumping function and relevant rhythm risk.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

3



Cancer imaging

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine computed tomography (CT), positron emission tomography (PET) or tumour-marker testing is not the usual screening plan for someone without symptoms.

Assess response or follow-up needs.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

4

Bone health

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

Assess fracture risk where relevant.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan



STEP 1

Before the first visit

Confirm the active regimen, blood results, cardiac checks, medicines review, route and consent discussion.



STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Phase-specific visit pattern

Infusion and observation times differ by drug and first versus later doses; the unit confirms the full visit duration, including blood review and preparation.

When scans, heart checks and bloods happen

Agree who books each test, who reviews the result and what happens if a check is overdue.

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose. The selected chemotherapy can require extra or more frequent checks.

Heart ultrasound and electrocardiogram (ECG)

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Scans and tumour markers

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine CT, PET or tumour-marker testing is not the usual screening plan for someone without symptoms.

Bone health and endocrine treatment

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

How surgery and radiotherapy fit

The selected indication sets the order: neoadjuvant treatment → surgery → selected postoperative treatment; adjuvant treatment starts after surgery. Finish any anthracycline phase before starting these antibodies with the taxane. Radiation timing is agreed with the treating teams.

SACTCHECK · CHAPTER 03

Symptoms and supportive care

Pertuzumab + trastuzumab · with selected chemotherapy · NCCP 00350

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Antibodies · Day 1 every 21 days	 **Day 1** Selected chemotherapy has its own calendar
Heart / breathing symptoms	 **Days 1–21** Report at any time
Diarrhoea	 **Days 1–21** Report early

Planned treatment: IV pertuzumab and IV trastuzumab every 21 days. Before surgery: 3–6 cycles alongside the selected chemotherapy. After surgery: up to 18 cycles overall under the agreed adjuvant plan. If an anthracycline is used first, finish it before starting these antibodies with the taxane phase. Confirm the exact companion regimen and local access.

Next phase: The chosen chemotherapy, indication, response, tolerance and local access determine the next step. Do not treat this antibody framework as a complete chemotherapy protocol.

Heart checks: Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Why these bands stay broad: Treatment markers show the planned schedule. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00350 v3 · official regimen ↗

There are no guaranteed "safe days". Serious symptoms need assessment on any day, even if the chart shows a usual recovery period.

Between-treatment notes

Day 1–21: The current treatment cycle

IV pertuzumab and IV trastuzumab every 21 days. Before surgery: 3–6 cycles alongside the selected chemotherapy. After surgery: up to 18 cycles overall under the agreed adjuvant plan. If an anthracycline is used first, finish it before starting these antibodies with the taxane phase. Confirm the exact companion regimen and local access. Your team confirms treatment dates and checks. Serious symptoms need advice on any day.

Support around the medicines

1 **Choose the companion chemotherapy**

Confirm the exact chemotherapy regimen before issuing its anti-sickness or other supportive prescription. The minimal emetic risk of the two antibodies does not describe the whole combination.

2 **Keep heart checks on the plan**

An echo every 12 weeks is separate from blood tests or cancer imaging. Ask who books and reviews it.

3 **Plan for symptoms**

Keep your agreed contact numbers and instructions. New symptoms can need assessment before a routine appointment.

Questions worth asking

Who arranges my tests?

Ask the team who books each test, who checks the result and who to contact if an appointment is overdue. This map does not book appointments.

What if I become unwell between visits?

Use the urgent contact plan for sudden illness, fever, chest symptoms, severe diarrhoea or poor fluid intake. Do not wait for a scheduled blood test or scan.

Try explaining the plan back: Which medicines are active now, which days do you receive or take them, and who is booking and reviewing your next tests?

Pertuzumab

- Heart pumping-function reduction
- Diarrhoea and rash
- Fatigue

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Combination visit · select the chemotherapy

Active medicines: Pertuzumab + Trastuzumab + Separately selected chemotherapy

Emetogenic risk: Determined by the selected chemotherapy

Chemotherapy has not been specified here. A complete supportive prescription cannot be generated from this antibody framework alone.

Additional phase-specific support

- Open the exact companion NCCP regimen before prescribing. Its antiemetics, premedication, granulocyte colony-stimulating factor (white-cell growth support) (G-CSF), blood thresholds and phase schedule apply.
- Antibody-specific premedication is not usually needed; consider paracetamol/antihistamine for the reaction history.
- If an anthracycline phase is selected, finish it before starting these antibodies with the taxane.
- Confirm indication and current access: NCCP00350b marks adjuvant pertuzumab reimbursement as not approved.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Antibodies only · if separately prescribed

Active medicines: Pertuzumab + Trastuzumab

Emetogenic risk: Minimal for antibodies only

No routine prophylactic antiemetic order is populated for this phase. Agree an individual symptom-treatment plan where needed.

Additional phase-specific support

- Confirm that chemotherapy has actually ended or is separately held; do not apply this minimal-risk script to a combination visit.
- No routine cytotoxic antiemetic, taxane premedication or chemotherapy granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) is carried forward.
- Continue cardiac checks every 12 weeks and at completion; a phase change does not reset the clock.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

NCCP 00350 · VERSION 3

Pertuzumab + trastuzumab · with selected chemotherapy

IV pertuzumab and IV trastuzumab every 21 days. Before surgery: 3–6 cycles alongside the selected chemotherapy. After surgery: up to 18 cycles overall under the agreed adjuvant plan. If an anthracycline is used first, finish it before starting these antibodies with the taxane phase. Confirm the exact companion regimen and local access.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.

1 Heart / breathing

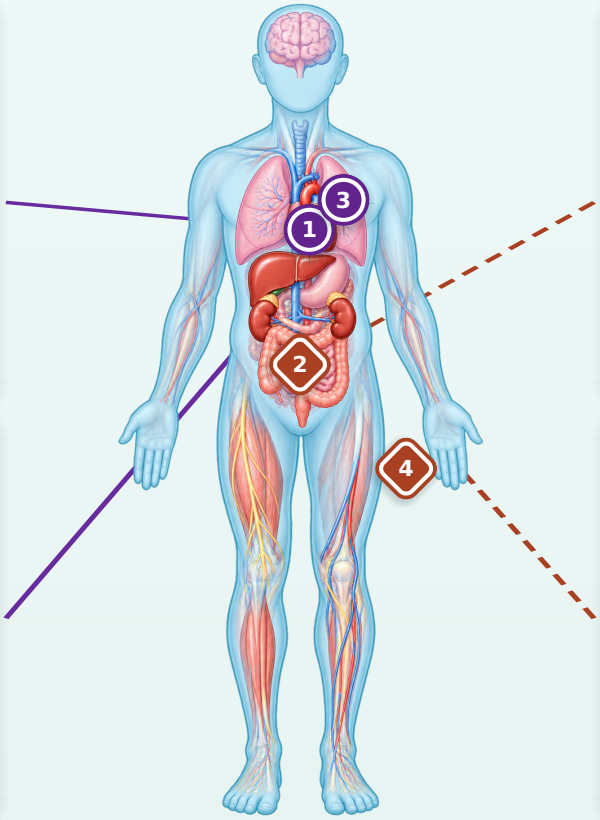
Pertuzumab + trastuzumab

Keep echo appointments; report breathlessness, swelling or reduced exercise tolerance.

2 Diarrhoea

Especially pertuzumab

Report watery stools early, especially with dizziness or poor intake.



3 Infusion reaction

Either antibody

Tell staff immediately about rash, chest tightness or breathing difficulty.

4 Rash / skin

HER2 antibodies

Report persistent or severe skin changes.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

This map covers the two HER2 antibodies only. The selected chemotherapy has additional risks and requires its own patient guide.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

SACTCHECK · CHAPTER 04

Your record and contact card

Pertuzumab + trastuzumab · with selected chemotherapy · National Cancer Control Programme (NCCP) 00350

Record each cycle separately. Each medicine has its own symptom table. Mark the worst symptoms since your last review, when they occurred and what they stopped you doing. Overlapping symptoms can be recorded under more than one medicine; you do not have to identify the cause.

These are patient-language severity prompts, not a validated questionnaire or a formal Common Terminology Criteria for Adverse Events (CTCAE) grading tool. Your team assigns the clinical grade. Serious symptoms override the columns: call without waiting to decide a severity level.

Use a new record sheet for each cycle or review. Copy or print another sheet when needed; the number of sheets does not prescribe the length of treatment.

Cycle number: _____ Start date: _____ Next review: _____

Current regimen · reusable cycle record · copy this sheet for another cycle of the same phase.

Pertuzumab · HER2-targeted antibody

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Diarrhoea	□ Usual pattern	□ Up to 3 extra stools above normal	□ 4–6 extra stools/ day or at night: call	□ 7+ extra stools/ day or dehydration: urgent review	□ Collapse / severe dehydration: 112/999
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999

Trastuzumab · HER2-targeted antibody

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999
Diarrhoea	□ Usual pattern	□ Up to 3 extra stools above normal	□ 4–6 extra stools/ day or at night: call	□ 7+ extra stools/ day or dehydration: urgent review	□ Collapse / severe dehydration: 112/999

Dates / doses missed or changed / symptoms affecting daily life / questions:

Red flags override the table. Fever, shivers, suddenly unwell, chest symptoms or inability to drink: seek prompt advice. Severe breathing difficulty, collapse or severe chest pain: 112 / 999.

Carry your treatment details

Print, cut out and fold. Follow the contact plan from your own hospital.

SACTCHECK · SACT TREATMENT

Pertuzumab + trastuzumab · with selected chemotherapy

NCCP 00350 v3 · 21-day cycle

IV pertuzumab and IV trastuzumab every 21 days. Before surgery: 3–6 cycles alongside the selected chemotherapy. After surgery: up to 18 cycles overall under the agreed adjuvant plan. If an anthracycline is used first, finish it before starting these antibodies with the taxane phase. Confirm the exact companion regimen and local access.

1	2	3	4	5	6

Your contact plan

Oncology team: Insert numbers here
Out of hours: Insert numbers here.

Evenings / overnight: instructed out-of-hours pathway or nearest ED.
Emergency: 112 / 999.

Other hospital / own contact: _____

QR: https://sactcheck.com/patient/00350/ · no patient information.

Draft for review · SACTCheck v0.84.16

Sources and further information

Pertuzumab + trastuzumab · with selected chemotherapy · National Cancer Control Programme (NCCP) 00350

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00350 v3 · official regimen**
   https://healthservice.hse.ie/documents/6813/350_v3_Pertuzumab_Trastuz_Chemo.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **EMA Herceptin product information**
   https://www.ema.europa.eu/en/documents/product-information/herceptin-epar-product-information_en.pdf
7. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
8. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
9. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
10. **NOGG · section 7, aromatase inhibitors and fracture risk**
    https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
11. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
12. **ASCO · biomarkers in metastatic breast cancer**
    https://ascopubs.org/doi/10.1200/JCO.2015.61.1459
13. **EMA Perjeta product information**
    https://www.ema.europa.eu/en/documents/product-information/perjeta-epar-product-information_en.pdf